

Ahmed Arfaoui

AI/ML Engineer | Agentic AI, RAG & Production ML Systems

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🌐 LinkedIn 🐙 GitHub 📁 Portfolio

PROFESSIONAL SUMMARY

AI/ML Engineer building LLM-powered applications, RAG pipelines, multi-agent workflows, and predictive ML systems. Experience spans LangGraph orchestration, hybrid retrieval, FastAPI services, and ML evaluation, deployment, and monitoring. Software Engineering graduate with a Data Science & AI specialization.

PROFESSIONAL EXPERIENCE

AI & Agentic Systems Engineering Intern — Soft Stars

Dec 2025 – Jun 2026

Open Web Catcher — Multi-Agent Web Intelligence System

- Built a multi-agent AI pipeline using LangGraph and LangChain for autonomous browser interaction, classification, network inspection, evidence extraction, and structured persistence.
- Built conditional agent routing, tool-calling workflows, recovery strategies, and browser automation for dynamic websites where deterministic automation frequently failed.
- Evaluated multiple LLMs across **126 end-to-end runs**, tracking token usage, latency, tool calls, errors, and cost; recorded a **97.6% tool-call success rate** and an average persisted-run cost of **\$0.152**.
- Integrated PostgreSQL and Azure Service Bus for traceability, asynchronous processing, and reliable data flow.

Data Science Intern — VERMEG

Aug 2025 – Sep 2025

AI-Assisted Legacy Code Modernization

- Automated conversion of **50+ legacy XML security configurations** into Java classes using Spring Boot and Spring AI; achieved **90% conversion accuracy** and reduced manual refactoring effort by approximately **95%**.

Data Science Intern — ESPRIT

Jun 2025 – Aug 2025

RAG and Machine Learning Platform

- Developed FastAPI backend services, data models, and database infrastructure for a cross-platform education product; integrated Gemini-based RAG for personalized quiz generation.
- Evaluated 6 clustering approaches for automated group formation, improving silhouette score by **15%** while maintaining **sub-200 ms** inference latency.

BI & Data Science Intern — CMR Tunisie

Jul 2024 – Aug 2024

Analytics and Time-Series Forecasting

- Built data ingestion, cleaning, transformation, analytics, and forecasting workflows from multi-source sales data and compared ARIMA, SARIMA, SARIMAX, and Prophet models for sales planning.

SELECTED AI & ML PROJECTS

SignalRank — Hybrid Retrieval & Reranking

2026

Built a hybrid retrieval pipeline combining BM25, pgvector, reciprocal-rank fusion, and cross-encoder reranking; exposed retrieval through FastAPI/PostgreSQL and evaluated with Precision@K, Recall@K, MRR, and nDCG.

HuntFlow — Multi-Agent AI Platform

2026

Independently built an AI-native application coordinating specialized agents for analysis, semantic retrieval, document generation, and workflow automation using LangGraph orchestration, multi-LLM routing, fallbacks, and cost-aware execution.

PitWall ML — ML Engineering & MLOps

2026

Built an end-to-end predictive system using LightGBM, FastAPI, temporal validation, MLflow, Evidently, Prometheus/Grafana, Docker, and CI/CD; reduced prediction MAE from **8.44 s** to **1.64 s**.

TECHNICAL SKILLS

Generative AI & Agents — LLMs, RAG, AI Agents, Multi-Agent Systems, LangGraph, LangChain, tool calling, prompt engineering, semantic retrieval, agent evaluation, LLM evaluation

Machine Learning — Python, scikit-learn, LightGBM, PyTorch, TensorFlow/Keras, NLP, clustering, time-series forecasting, model evaluation, temporal validation

Retrieval & Data — PostgreSQL, SQL, pgvector, BM25, FAISS, cross-encoder reranking, reciprocal-rank fusion, Pandas, NumPy, ETL pipelines

Backend & APIs — FastAPI, REST APIs, WebSockets, asynchronous processing, Java, Spring Boot

MLOps & Cloud — Docker, GitHub Actions, CI/CD, MLflow, DVC, Evidently, Prometheus, Grafana, Azure Service Bus, AWS, Linux, Git

EDUCATION

ESPRIT — Engineering Degree in Software Engineering

Sep 2022 – Aug 2026

Data Science & AI Specialization | Highest Honors (Mention Excellent)

IPEIB — Preparatory Cycle for Engineering Schools

Sep 2019 – Jun 2022

Mathematics & Physics

ADDITIONAL

Certifications — NVIDIA: Building Transformer-Based NLP Applications | AWS Academy Cloud Foundations | Hashgraph Developer Course

Languages — Arabic: Native | French: Fluent | English: Fluent